

Executive Summary: Data-Driven Decision Support for the German Energiewende

Overview and Business Context Germany's energy transition (*Energiewende*) generates a massive, highly fragmented volume of high-frequency data, which frequently overwhelms the cognitive capacity of human decision-makers. This phenomenon, known as Bounded Rationality, forces energy professionals to rely on intuitive, reactive "System 1" thinking when attempting to manage the complex, conflicting goals of the Energy Trilemma (Energy Security, Energy Equity, and Environmental Sustainability).

To systematically solve this cognitive bottleneck, this study developed and empirically validated a macro-level Business Intelligence (BI) dashboard. By centralizing raw SMARD datasets (2019–2024) and integrating an ETS time-series forecasting algorithm, the prototype effectively transitions energy policy evaluation from a reactive reporting process into a deliberate, analytical scenario-planning tool.

Methodology The research utilized strict Convergent Mixed Methods design. The quantitative component successfully integrated and harmonized official SMARD datasets using a relational Star Schema architecture to ensure zero computational latency. The qualitative component validated the dashboard's strategic utility through a purposive survey of seven elite energy sector experts, strictly representing two critical professional domains: macro-level policy planning strategists and sustainability/renewable integration specialists.

Key Strategic Findings

1. The 2030 Policy Gap and Statutory Risks While the quantitative data confirms a monumental milestone, renewable energy generation has successfully eclipsed legacy fossil baseloads to cover 50.3% of total gross consumption; the predictive ETS forecasting model exposes a severe structural vulnerability. The model identifies a distinct **25% to 30% "Policy Gap"** between the current linear trajectory of renewable integration and the strict legal mandate of an 80% renewable share by 2030. This visually proves that the current infrastructure deployment rate is structurally insufficient.

2. Wholesale Market Volatility and The Merit Order Effect The dashboard provides explicit empirical evidence of the "Merit Order Effect" and severe economic friction within the German wholesale market. During periods of low weather-dependent renewable generation, the "Residual Load" spikes, forcing the grid to immediately dispatch expensive fossil fuels (hard coal and natural gas) to prevent blackouts. Because these fossil fuels dictate the final marginal market price, **the physical inflexibility of the grid is the primary driver of extreme wholesale price volatility.** Conversely, the dashboard recorded severe market inefficiency during renewable oversupply periods, exposing 210 instances of negative prices in a single month (May).

3. The Dual-Value Structure of Business Intelligence The expert validation survey confirmed that the BI dashboard delivers two distinct, highly significant forms of value across different professional workflows:

- **Operational Efficiency for Analysts:** By automating the initial data-gathering phase, the system frees experts from unproductive administrative tasks. Survey data confirms that **85.7% of experts saved between 1 and 4 hours per week.**
- **Strategic Clarity for Executives:** For senior strategists operating at a macro-policy level, operational time savings are not the primary metric. Instead, the tool delivers **absolute**

decision clarity and proactive "Situational Awareness". The expert panel achieved a perfect consensus (Median = 4.0) that the unified interface significantly reduces the cognitive load required to process complex Trilemma trade-offs.

4. Algorithmic Transparency in Predictive Governance The study established a critical boundary condition for deploying AI and predictive BI tools in strict regulatory environments. While the predictive visualization successfully improved perceived agility and proactive planning (Median = 4.0), the interpretability of those forecasts scored neutrally (Median = 3.0) because experts actively withheld their trust without explicit statistical parameters. The study concludes that algorithms only successfully shift decision-makers toward analytical evaluation when uncertainty bounds (such as labeled 95% confidence intervals) are made fully transparent. This necessitates **"algorithmic transparency design"** as a mandatory phase in future BI development.

Recommendations for Policy and Industry Strategy

1. **Accelerate Parallel Investments in Grid Flexibility:** Expanding wind and solar capacity is no longer sufficient; to stabilize day-ahead prices and protect consumers from economic exposure, there must be massive, incentivized investments in utility-scale storage infrastructure (e.g., Pumped Storage, Power-to-Gas) to absorb renewable oversupply and discharge during peak demand.
2. **Institutionalize Predictive Business Intelligence:** Regulatory authorities and corporate strategy departments must transition from static, historical reporting to dynamic, predictive analytics. Institutionalizing tools that visually project mathematical confidence intervals prevent reliance on illusory grid stability and enforces proactive scenario planning.
3. **Integrate Targeted Carbon KPIs:** Future strategic frameworks should directly integrate real-time CO₂ Emission Intensity metrics (gCO₂/kWh) to explicitly track exact carbon reductions alongside wholesale market prices, bridging the gap between environmental goals and economic realities.